Unofficial worked solution — written by UWMathFinder, not by the instructor. Spot an error? Report it on the site.

Math 208 Final, Spring 2025 — Problem 1

1. $R_1 \to \frac{1}{2}R_1$: $(1, 0, 4, 0, 2)$.
2. $R_2 \to R_2 + R_1$: $(0, 1, 3, 1, x+2)$.
3. $R_3 \to R_3 - R_2$: $(0, 0, 0, 1, 0)$.
4. $R_2 \to R_2 - R_3$: $(0, 1, 3, 0, x+2)$.

$$\mathrm{RREF} = \boxed{\begin{pmatrix} 1 & 0 & 4 & 0 & 2 \\ 0 & 1 & 3 & 0 & x+2 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}}.$$

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Math 208 Final, Spring 2025 — Problem 2

Yes. $\vec{v}$ and $\vec{w}$ are independent (not multiples of each other). Add $\vec{e}_3$ and $\vec{e}_4$:

$$\det \begin{bmatrix} 1 & 1 & 0 & 0 \\ 3 & 0 & 0 & 0 \\ 2 & 2 & 1 & 0 \\ 0 & 2 & 0 & 1 \end{bmatrix} = -3 \neq 0,$$

so $\{\vec{v}, \vec{w}, \vec{e}_3, \vec{e}_4\}$ is linearly independent, and four independent vectors in $\mathbb{R}^4$ form a basis.

Math 208 Final, Spring 2025 — Problem 3

$\det B = 2(4-1) - 1(2-1) + 1(1-2) = 4 \neq 0$, so B is invertible. Row reducing $[B \mid I]$ (or using the adjugate):

$$\boxed{B^{-1} = \frac{1}{4}\begin{pmatrix} 3 & -1 & -1 \\ -1 & 3 & -1 \\ -1 & -1 & 3 \end{pmatrix}}.$$

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Math 208 Final, Spring 2025 — Problem 4

Expand repeatedly along columns with a single nonzero entry. Column 1 gives -1; then the $(2,2)$ entry gives -41; then the row of the remaining 3×3 block $\begin{pmatrix} 3 & 1 & 6 \\ 0 & 2 & 0 \\ 2 & 2 & -1 \end{pmatrix}$ with a single entry gives 2; what's left is $\begin{pmatrix} 3 & 6 \\ 2 & -1 \end{pmatrix}$ with $\lambda^2 - 2\lambda - 15 = (\lambda - 5)(\lambda + 3)$.

$$\boxed{\lambda = -1,\ -41,\ 2,\ 5,\ -3}.$$

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Math 208 Final, Spring 2025 — Problem 5

$S = \text{null}(D^T)$: $x_1 + 3x_2 = 0$ and $x_1 + 3x_3 + 6x_4 = 0$. With x_3, x_4 free, $x_1 = -3x_3 - 6x_4$, $x_2 = x_3 + 2x_4$:

$$S = \text{span}\{(-3, 1, 1, 0),\ (-6, 2, 0, 1)\} \quad \text{(both dot to 0 with each column)}.$$

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Math 208 Final, Spring 2025 — Problem 7

The matrix of T (in the order w, x, y, z) is

$$\begin{bmatrix} 5 & -1 & 0 & 0 \\ 2 & 0 & 0 & 0 \\ 7 & 2 & 1 & 0 \\ 13 & 41 & 11 & 4 \end{bmatrix}, \quad \det = 8 \neq 0.$$

It is invertible, so $\ker T = \{\vec{0}\}$ and $\boxed{\dim \ker T = 0}$. (Directly: $2w = 0 \Rightarrow w = 0$, then $x = 0$, $y = 0$, $z = 0$.)

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Math 208 Final, Spring 2025 — Problem 8

$G = ABA^{-1}$ is similar to B: same eigenvalues $\boxed{2 \text{ (twice)},\ -1}$. If $B\vec{u} = \lambda\vec{u}$ then $G(A\vec{u}) = AB\vec{u} = \lambda A\vec{u}$, so the eigenvectors of G are A times those of B (the columns of A):

$$E_2 = \text{span}\{(1,2,-1),\ (0,1,3)\}, \qquad E_{-1} = \text{span}\{(0,0,1)\}.$$

Unofficial worked solution — written by UWMathFinder, not by the instructor. Spot an error? Report it on the site.

Math 208 Final, Spring 2025 — Problem 9

Eigenvalues $0, 1, 2$ (here A means F). Row reducing $F - \lambda I$:

$$E_0 = \text{span}\{(1, -1, 3)\}, \quad E_1 = \text{span}\{(1, -1, 4)\}, \quad E_2 = \text{span}\{(2, -1, 5)\}.$$

(Check: $F(1, -1, 3) = \vec{0}$, $F(1, -1, 4) = (1, -1, 4)$, $F(2, -1, 5) = (4, -2, 10)$.)

Unofficial worked solution — written by UWMathFinder, not by the instructor. Spot an error? Report it on the site.

Math 208 Final, Spring 2025 — Problem 10

$AB = \begin{pmatrix} -1 & 1 \\ 0 & -1 \end{pmatrix} = -\begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$. So T is a horizontal shear $(x, y) \mapsto (x - y, y)$ followed by a rotation by $180°$ (point reflection through the origin). The face is turned upside down and slanted: points with $y > 0$ are pushed left before the rotation, so the image's top half leans right, and the eyes end up below the center with the smile on top. The origin stays fixed and $(1, 0) \mapsto (-1, 0)$.

Bonus. $\det A(x) = x^3 - 3x + 2 = (x-1)^2(x+2)$, so $A(x)$ is invertible for $\boxed{x \neq 1,\ -2}$. Writing $A(x) = (x-1)I + J$ with J the all-ones matrix ($J^2 = 3J$):

$$A(x)^{-1} = \boxed{\frac{1}{x-1}\Big(I - \frac{1}{x+2}J\Big)}.$$